

StyleGAN3: Generative Networks for Improving the Equivariance of Translation and Rotation

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Abstract - StyleGAN can use style to affect facial posture and identity features, and noise to affect hair, wrinkles, skin color and other details. Among these, the outcomes of the picture processing will vary slightly between different versions of styleGAN. As a result, the comparison of performance differences between styleGAN2 and the two modified versions of styleGAN3 will be the main focus of this study. We used the FFHQ dataset as the dataset and FID, EQ-T, and EQ-R were used to be the assessment of the model. In the end, we discovered that Stylegan3 version is a better generative network to improve the equivariance. Our findings have a positive impact on the creation of animation and videos.

Index Terms – StyleGAN2, StyleGAN3, Assessment, Equivariance.

I. INTRODUCTION

In both video and still images, the image's sharpness is crucial because it not only makes for better viewing but also advances technology. To enhance the image's quality, people are constantly working to create more applicable models. Among them, styleGAN is a significant model that can produce photographs of a high caliber. The image quality can be continuously adjusted with the development and advancement of various models.

We compare the performance variations between styleGAN2 and styleGAN3 in producing graphics in this study. StyleGAN2 is an adversarial network made up of a generator and a discriminator that produces excellent facial images. Image details are under the control of the generator, which also extracts image features. StyleGAN3 performs comparable tasks to styleGAN2, however the distinction is that styleGAN3 can enhance the network's building and training processes. We choose FFHQ [10] as the data set since it contains a variety of images that can improve the results' accuracy. As assessment indices for contrasting styleGAN2 with styleGAN3, FID [2], EQ-T [8], and EQ-R are chosen. By comparing these evaluation indexes, we can clearly conclude that the performance of the fine-tuning model is obviously better than that of the original model.

The results revealed that the photos could be presented appropriately and that simple details could be shown clearly after we applied the debugged and improved model to more photographs.

In conclusion, our enhanced fine-tuning model performs better than the original model. In addition to advancing face and object identification technology, this will directly enhance image and video clarity to suit a variety of requirements.

II. LITERATURE REVIEW

In generating pair models, styleGAN is an important model that can generate high-quality images. In the development of styleGAN, styleGAN2 and styleGAN3 are two essential versions. This article will compare the two versions and analyze their performance differences. First of all, in terms of the quality of generated images, styleGAN3 is significantly better than styleGAN2 [1]. When evaluating the quality of generated images, there are also significant differences between styleGAN3 and styleGAN2. Compared with styleGAN2, styleGAN3 has significantly improved the FID (Fréchet Perception Distance) evaluation index [2], which shows that the image generated by styleGAN3 is more realistic [3]. In addition, styleGAN3 is also better for evaluating adversarial networks [4]. Secondly, styleGAN3 is more efficient in the training process. Regarding training time, styleGAN3 is about half shorter than styleGAN2 and uses fewer parameters during training [6], which means that styleGAN3 can train on smaller data sets [7].

To sum up, styleGAN3 is significantly better than styleGAN2 in performance. It is more efficient in generating image quality and training processes. Therefore, styleGAN3 is a better choice for practical applications.

III. METHODOLOGY AND TRAINING METHOD

A. Methodology

1. StyleGAN2

StyleGAN3: 用于提升平移和旋转等变性的生成网络

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Abstract - StyleGAN 可以利用风格来影响面部姿态和身份特征, 而噪声则会影响头发、皱纹、肤色等细节。在这些因素中, 不同版本的StyleGAN处理图片的结果会略有差异。因此, 本研究主要关注styleGAN2与两个修改版的styleGAN3之间的性能差异比较。我们使用了FFHQ数据集作为数据源, 并采用FID、EQ-T和EQ-R来评估模型。最终发现, StyleGAN3版本是提高等变性的更好生成网络。我们的研究成果对动画和视频创作具有积极影响。

Index Terms – *StyleGAN2, StyleGAN3, Assessment, Equivariance.*

I. 引言

在视频和静态图像中, 图像的锐度至关重要, 因为它不仅提升了观看体验, 还推动了技术进步。为了提升图像质量, 人们不断努力开发更实用的模型。其中, StyleGAN是一个重要的模型, 能够生成高质量的照片。随着各种模型的发展与改进, 可以持续调整图像的质量。

我们在这项研究中比较了styleGAN2和styleGAN3在生成图形时的性能差异。StyleGAN2是一个由生成器和判别器组成的对抗网络, 能够产生高质量的人脸图像。图像细节受生成器控制, 并且生成器还提取图像特征。StyleGAN3执行与StyleGAN2类似的任务, 但区别在于StyleGAN3可以增强网络构建和训练过程。我们选择FFHQ数据集[10]作为研究对象, 因为它包含多种类型的图片, 有助于提高结果的准确性。为了对比styleGAN2和styleGAN3, 我们选择了FID[2]、EQ-T[8]和EQ-R作为评估指标。通过比较这些评价指标, 我们可以清楚地得出结论: 微调模型的表现明显优于原始模型。

结果显示, 在应用调试并改进后的模型后, 照片能够被适当呈现, 并且简单细节可以清晰展示。

总之, 我们的改进型微调模型优于原模型。除了推进人脸和物体识别技术的发展之外, 这还将直接提升图像和视频的清晰度, 以满足各种需求。

第二章 文献回顾

在生成配对模型时, StyleGAN是一个重要的模型, 能够生成高质量的图像。在StyleGAN的发展过程中, StyleGAN2和StyleGAN3是两个关键版本。本文将比较这两个版本, 并分析它们的性能差异。首先, 在生成图像的质量方面, StyleGAN3显著优于StyleGAN2[1]。在评估生成图像质量时, StyleGAN3与StyleGAN2之间也存在显著差异。相较于StyleGAN2, StyleGAN3大幅提升了FID (Fréchet感知距离) 评价指标[2], 表明由StyleGAN3生成的图像更加逼真[3]。此外, StyleGAN3在评估对抗网络方面表现更佳[4]。其次, 在训练过程中, StyleGAN3更为高效。就训练时间而言, StyleGAN3比StyleGAN2缩短了大约一半, 并且在训练时使用的参数较少[6], 这意味着StyleGAN3可以在较小的数据集上进行训练[7]。

总之, StyleGAN3 在性能上显著优于 StyleGAN2。它在生成图像质量和训练过程方面更为高效。因此, StyleGAN3是实际应用中的更好选择。

第三章 方法论与训练方法

A. Methodology

风格生成对抗网络2 (StyleGAN2)

StyleGAN2 is a generation antagonism network (GAN) used to generate high-quality face images. It consists of two main parts: generator and discriminator.

Generator: The generator adopts an architecture called "style based". It uses a set of trainable parameters (called "style") to control the details and appearance of the image.

Discriminator: The discriminator is responsible for recognizing the actual image and generating the image. It uses a convolution neural network (CNN) to extract image features and then uses a complete connection layer to output whether the image is actual or generated.

StyleGAN2 can generate high-quality face images through training generators and discriminators.

2. StyleGAN3

StyleGAN3 is a generation countermeasure network (GAN) that generates high-quality images. Compared with StyleGAN2, StyleGAN3 has some improved network structure and training methods.

Generator: Unlike StyleGAN2, StyleGAN3 uses deeper networks and more complex network layers, such as the bottleneck layer and nonlinear convolution layer. This enables StyleGAN3 to generate higher-quality images.

Discriminator: The structure of the discriminator is similar to that of the StyleGAN2 discriminator.

Training method: Unlike StyleGAN2, StyleGAN3 adopts a new training method called "path length regulation." This method improves the generated image's quality by limiting the generator's length in the style space. The formula is as follows:

$$L_{reg} = \|w_{i+1} - w_i\|_2^2 \quad (1)$$

Where w_i and w_{i+1} is the continuous point of the style parameter.

In short, StyleGAN3 is an advanced generation antagonism network. Its generated image quality is higher than that of StyleGAN2, and new training methods have been adopted to improve the image quality.

B. Evaluation indicator

1. FID:

FID (Fréchet Perception Distance) is a measure used to evaluate the quality of generated images. It evaluates the image quality by calculating the distance between the actual image and the generated image in the Perception-v3 network feature space. The formula is as follows:

$$FID(X, G(z)) = \|\mu_x - \mu_G\|^2 + Tr(\Sigma_x + \Sigma_G - 2(\Sigma_x \Sigma_G)^{\frac{1}{2}}) \quad (2)$$

Where x represents the actual image, $G(z)$ represents the generated image, μ and Σ They represent the mean and covariance in the Inception-v3 network feature space.

2. EQ-T:

EQ-T (Quality of Distribution) is a measure used to evaluate the quality of generated images [8]. It evaluates whether the distribution of generated images is the same as that of natural images. EQ-T evaluates the image quality by calculating the JS (Jensen-Shannon) divergence of the actual image and the generated image in the Perception-v3 network feature space.

3. EQ-R:

EQ-R (Quality of Representation) is a measure used to evaluate the quality of generated images. It evaluates whether the generated images can find similar representations to authentic images in the Inception-v3 network feature space. EQ-R evaluates the image quality by calculating the L2 distance of the actual image and the generated image in the Inception-v3 network feature space.

These three methods of evaluating image quality are based on the feature space of the Perception-v3 network. In contrast, FID evaluates images based on the distance between the feature spaces of the two images.

IV. EXPERIMENTS

We implemented our frameworks in pytorch [9] with RTX 3090 GPU, conducting extensive experiments on FFHQ [10] and Wombo Dream [11] based on styleGAN2 [12] and styleGAN3 [13]. Then we will use the common evaluation metrics FID [14] and EQ.

A. FFHQ dataset

The full name of FFHQ is Flickr-Faces-High-Quality (Flickr-Faces-HQ). It was originally created as a benchmark for Generative Adversarial Networks (GAN) and is also used in the training data set of Style GAN and was open sourced by Nvidia in 2019. [10] FFHQ is a high-quality face data set, which contains 70,000 high-definition face images in PNG format with a resolution of 1024×1024. It is rich and diverse in age, race, and image background, and has obvious differences in face attributes, so it can also be used to develop some face attribute classification or face semantic segmentation models.

B. StyleGAN2

This model mainly uses Adaptive Instance Normalization (AdaIN) [18] to quickly realize the conversion of any feature map and achieves the purpose of conversion through the mean and variance of the style graph. [15] However, it will lead to the destruction of information between features brought about by normalization, so we will introduce the styleGAN2 model [12], cancel the mean when normalizing features, and add noise outside the style module. Of course, we will also use methods such as weight demodulation, lazy regularization, and path length regularization to improve the performance of the model.

C. StyleGAN3

We must admit that when using the GAN [16] model to generate pictures, the model features will be too dependent on absolute pixel coordinates, which directly leads to the problem

StyleGAN2 是一种生成对抗网络（GAN），用于生成高质量的人脸图像。它主要由生成器和判别器两部分组成。

生成器采用了一种称为“基于风格”的架构。它使用一组可训练的参数（称为“风格”）来控制图像的细节和外观。

判别器：判别器负责识别真实图像和生成的图像。它使用卷积神经网络（CNN）提取图像特征，然后通过全连接层输出该图像是真实的还是生成的。

StyleGAN2 可以通过训练生成器和判别器来生成高质量的人脸图像。

2. StyleGAN3

StyleGAN3 是一种生成对抗网络（GAN），能够生成高质量的图像。与 StyleGAN2 相比，StyleGAN3 在网络结构和训练方法上有所改进。

生成器：与 StyleGAN2 不同，StyleGAN3 使用更深的网络和更复杂的网络层，例如瓶颈层和非线性卷积层。这使得 StyleGAN3 能够生成更高质量的图像。

判别器：判别器的结构与 StyleGAN2 的判别器类似。

训练方法：与 StyleGAN2 不同，StyleGAN3 采用了一种称为“路径长度调节”的新训练方法。该方法通过限制生成器在风格空间中的长度来提高生成图像的质量。公式如下：

$$L_{reg} = \|w_{i+1} - w_i\|_2^2 \quad (1)$$

其中 w_i 和 w_{i+1} 是风格参数的连续点。

简而言之，StyleGAN3 是一种先进的生成对抗网络。其生成的图像质量高于 StyleGAN2，并且采用了新的训练方法来提升图像质量。

B. Evaluation indicator

1. FID:

FID（Fréchet感知距离）是一种用于评估生成图像质量的度量。它通过计算实际图像与生成图像在Perception-v3网络中的距离来评价图像质量。

特征空间。公式如下：

$$FID(X, G(z)) = \|\mu_x - \mu_G\|^2 + \text{Tr}(\Sigma_x + \Sigma_G - 2(\Sigma_x \Sigma_G)^{\frac{1}{2}}) \quad (2)$$

其中 x 表示实际图像， $G(z)$ 表示生成的图像， μ 和 Σ 分别表示 Inception-v3 网络特征空间中的均值和协方差。

2. EQ-T:

EQ-T（分布质量）是一种用于评估生成图像质量的度量标准[8]。它判断生成的图像分布是否与自然图像的分布相同。通过计算感知网络v3特征空间中实际图像和生成图像之间的JS（Jensen-Shannon）散度，来衡量图像的质量。

3. 等效队列重排序（EQ-R）：

EQ-R（表示质量）用于评估生成图像的质量。它衡量生成的图像是否能在Inception-v3网络特征空间中找到与真实图像相似的表示。通过计算实际图像和生成图像在Inception-v3网络特征空间中的L2距离来评价图像质量。

这三种图像质量评估方法基于Perception-v3网络的特征空间。相比之下，FID是根据两张图像的特征空间之间的距离来评估图像的质量。

第四节 实验

我们使用 PyTorch [9] 在 RTX 3090 GPU 上实现了我们的框架，并在基于 styleGAN2 [12] 和 styleGAN3 [13] 的 FFHQ [10] 和 Wombo Dream [11] 数据集上进行了广泛的实验。然后我们将使用常用的评估指标 FID [14] 和 EQ 进行评价。

A. FFHQ dataset

FFHQ 的全称是 Flickr-Faces-High-Quality（Flickr-Faces-HQ）。它最初作为生成对抗网络（GAN）的基准测试而创建，并且也被用于 Style GAN 的训练数据集，该数据集由 Nvidia 在 2019 年开源。[10] FFHQ 是一个高质量的人脸数据集，包含 70,000 张分辨率为 1024×1024 的高清晰度 PNG 格式人脸图像。这些图像在年龄、种族和背景方面丰富多样，并且面部属性差异明显，因此也可以用于开发一些面部属性分类或面部语义分割模型。

B. StyleGAN2

该模型主要采用自适应实例归一化（AdaIN）[18]来快速实现任意特征图的转换，并通过风格图的均值和方差达到转换的目的。然而，这会导致归一化过程中特征间信息的破坏，因此我们将引入StyleGAN2模型[12]，在归一化特征时取消均值计算，并在外加噪声以增强风格模块的效果。当然，我们还将采用权重重解调、懒惰正则化和路径长度正则化等方法来提升模型性能。

C. StyleGAN3

我们必须承认，在使用GAN[16]模型生成图片时，模型特征会过于依赖绝对像素坐标，这直接导致了问题。

of sticking detailed features [17]. And styleGAN3 fundamentally solves this problem, realizes the invariance of image translation and rotation, and greatly improves the quality of image synthesis [13]. Figure below shows the qualitative comparison of image generation. You will find that, after comparison, the hair of the animal does not stick.



Figure 1 Comparison of the effect of StyleGAN3 technological innovation

Translation invariance [13]: These layers tend to have rich frequency information near their bandwidth limits, which requires very strong attenuation to eliminate aliasing. So, frequency hopping is higher in the lowest resolution layer to maximize the attenuation of the stopband, but lower in the highest resolution layer to match the high frequency detail.

Rotation invariance [13]: Replace the convolution kernel size for all layers from 3*3 to 1*1 (a convolution operation based on 1*1 can produce a strong rotating isomorphic generator, which can force the model to achieve a more natural hierarchical refinement once aliasing is properly suppressed); The number of feature maps is doubled to make up for the reduced feature capacity.

model	FID	EQ-T	EQ-R
styleGAN2	5.14	-	-
+Filtered nonlinearities	6.35	30.60	10.81
+ Boundaries & up sampling	6.02	24.62	10.97
+Simplified generator	5.21	19.47	10.41
+Fourier features	4.79	16.23	10.81
+ Non-critical sampling	4.78	43.90	10.84
+ Transformed Fourier features	4.64	45.20	10.61
+ Flexible layers (StyleGAN3-T)	4.62	63.01	13.12
+ No noise inputs	4.54	15.81	10.84
+ Rotation equiv. (StyleGAN3-R)	4.50	66.65	40.48
styleGAN2 (Train from scratch)	3.79	15.89	10.79

Table 1 using FFHQ-U as the data set to conduct experiments to compare the performance of styleGAN2 and styleGAN3.

D. Training Stability

Since stylegan3 came out, we've been fine-tuning the model, experimenting with more data sets to compare the resulting images. To be specific, we mainly choose the models within styleGAN3-T [13] and styleGAN3-R [13]. Due to the limitation of RTX 3090, we set the batch value to 24, which also caused the need to increase the R1 RE weight, so gamma was selected as 10. Then we selected the number of iterations as 500 (kimg), because it was found in the experiment that the change of results would no longer affect the result after 500 times. To save power costs, we set metrics to none.

E. Result

model	FID	EQ-T	EQ-R
styleGAN2 (Train from scratch)	3.79	15.89	10.79
styleGAN3-T (ours)	3.67	61.78	14.01
styleGAN3-R (ours)	3.61	64.81	47.73

Table 2 using FFHQ-U as the data set based on the better performing models about styleGAN2 and styleGAN3.

After model transformation and upgrading, we selected the three most representative models for fine-tuning training. Through the above training strategies, we obtained the results of styleGAN3 model [13] based on FFHQ-U [10] data set. The performance of the fine-tuning model is obviously better than that of the original model.

F. Image generation



Figure 2 the original picture



Figure 3 after using styleGAN3-T (12 kimg)



Figure 4 after using styleGAN3-T (68 kimg)

After debugging and improving the model through the FFHQ dataset [10], we decided to study the general generality of the model and decided to try more data sets. All the source images used in this experiment were from Wombo Dream [11] and were cropped to centre and scaled to 1024x1024. In this experiment, StyleGan3-T model [13] was adopted. The images above show some samples of styleGAN3 [13] in action.

详细的粘连特征问题[17]。而StyleGAN3从根本上解决了这一问题，实现了图像变换和旋转的不变性，并大幅提升了图像合成的质量[13]。下图展示了图像生成的定性比较。你会发现，在对比之后，动物的毛发不再出现粘连现象。



图1 StyleGAN3 技术创新的效果对比

频率不变性[13]: 这些层在其带宽极限附近往往包含丰富的频域信息，这要求非常强的衰减来消除混叠现象。因此，在最低分辨率层中采用更高的频率跳变以最大化阻带内的衰减，而在最高分辨率层中则降低频率跳变以匹配高频细节。

旋转不变性 [13]: 将所有层的卷积核大小从 3×3 更改为 1×1 （基于 1×1 的卷积操作可以生成一个强大的旋转同构生成器，这可以在适当抑制混叠后迫使模型实现更自然的层次细化）；特征图的数量翻倍以弥补特征容量的减少。

model	FID	EQ-T	EQ-R
styleGAN2	5.14	-	-
+Filtered nonlinearities	6.35	30.60	10.81
+ Boundaries & up sampling	6.02	24.62	10.97
+Simplified generator	5.21	19.47	10.41
+Fourier features	4.79	16.23	10.81
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Table 1 using FFHQ-U as the data set to conduct experiments to compare the performance of styleGAN2 and styleGAN3.

D. Training Stability

自 StyleGAN3 发布以来，我们一直在对模型进行微调，并使用更多的数据集来比较生成的图像。具体来说，我们主要选择了 styleGAN3-T [13] 和 styleGAN3-R [13] 模型。由于 RTX 3090 的限制，我们将批处理大小设置为 24，这也导致了需要增加 R1 RE 权重，因此选择 γ 为 10。然后我们选择了迭代次数为 500 (kimg)，因为在实验中发现超过 500 次后结果的变化不再影响最终效果。为了节省电力成本，我们将度量标准设置为 none。

E. Result

model	FID	EQ-T	EQ-R
styleGAN2 (Train from scratch)	3.79	15.89	10.79
styleGAN3-T (ours)	3.67	61.78	14.01
styleGAN3-R (ours)	3.61	64.81	47.73

Table 2 using FFHQ-U as the data set based on the better performing models about styleGAN2 and styleGAN3.

经过模型转换和升级后，我们选择了三个最具代表性的模型进行微调训练。通过上述训练策略，我们基于FFHQ-U数据集获得了styleGAN3模型[13]的结果。微调后的模型性能明显优于原始模型。

F. Image generation



Figure 2 the original picture



Figure 3 after using styleGAN3-T (12 kimg)



图4 使用styleGAN3-T (68 kimg) 后的效果

在通过FFHQ数据集[10]进行调试和改进模型之后，我们决定研究该模型的泛化能力，并尝试更多数据集。本实验中使用的源图像均来自Wombo Dream[11]，并被裁剪至中心位置且缩放为1024x1024大小。在本次实验中采用了StyleGAN3-T模型[13]。上方展示了一些风格化生成器styleGAN3[13]的实际效果样本。

According to the model iteration of the same data set above, it can be found that the hierarchical structure can achieve accurate position and even simple details can be well adhered by the intermediate layer through image boundary, pixel noise input and position encoding and aliasing.

V. CONCLUSION AND DISCUSSION

We compared the influence brought by the improvement of style GAN model through experiments and compared the performance of the final model to generate pictures. However, the data set selected in this study is small and has not been truly popularized to all data sets. Faced with the training of large unstructured data sets like ImageNet or the expansion to a higher definition, the computational power and time cost are relatively large, so we cannot conduct experiments at present. Therefore, there is still a lot of room for improvement in the training strategy or model fine-tuning.

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根据上述数据集的模型迭代，可以发现层次结构能够实现准确的位置定位，并且通过图像边界、像素噪声输入和位置编码及混叠，中间层甚至能很好地处理简单的细节。

五. 结论与讨论

我们通过实验对比了改进后的风格GAN模型的影响，并评估了最终模型生成图片的性能。然而，本研究中选择的数据集较小，并未真正普及到所有数据集中。面对像ImageNet这样的大规模无结构化数据集训练或扩展至更高分辨率时，计算能力和时间成本相对较大，因此目前无法进行相关实验。因此，在训练策略或模型微调方面仍有很大的改进空间。

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请提供需要翻译的英文学术文本内容。根据您提供的指南，我将直接进行精准翻译，并保持原有段落结构。

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